HENIL PATEL

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EDUCATION

Master of Science in Data Science | ***Northeastern University, Boston*** Jan 2025 – *Expected* May 2027

WORK EXPERIENCE

Machine Learning Software Intern | ***Tech Rational*** Feb 2023 – May 2023

- Architected an end-to-end face recognition and automated attendance tracking system using OpenCV and Python, reducing **15%** administrative overhead.
- Trained a face recognition model using pre-trained **FaceNet** embeddings with rigorous data leakage prevention, ensuring reliable performance across variable lighting conditions.
- Built a real-time inference pipeline with MTCNN face detection and cosine similarity-based identity matching, **minimizing false-positive rates** in production.
- Implemented passive liveness detection via texture analysis to prevent spoofing, with face embeddings and records stored securely in **AWS S3** and **RDS** (MySQL).
- Packaged and deployed the system using **Flask REST API** and **Docker** on AWS **EC2**, with a Streamlit dashboard for real-time monitoring.

Machine Learning Research Assistant | ***Ganpat University*** Jul 2022 – Dec 2022

- Engineered a multi-stage NLP preprocessing pipeline on **100K+** DUC benchmark samples, integrating TF-IDF and semantic sentence embeddings via Sentence-BERT.
- Fine-tuned **BART** and **T5** summarization models benchmarked against TLBO, Firefly, and Cuckoo baselines, validated against human-written golden summaries.
- Implemented a RAG layer using **FAISS** vector indexing over DUC embeddings, reducing hallucinations and improving factual consistency during generation.
- Integrated **Mistral 7B** as a judge LLM to supplement ROUGE scoring with semantic coherence, factual consistency metrics.
- Deployed system as a **Streamlit** app containerized via **Docker** on AWS **EC2**, and co-authored a research paper published in the Journal of Electrical Systems.

PROJECTS

ChefBot: AI Recipe Generation Chatbot | *Gen AI*, *RAG*, *Prompt Engineering* Nov 2025 – Feb 2026

- Developed a full-stack AI recipe chatbot integrating **Llama 3.1 8B** LLM with a Streamlit UI, using **prompt engineering** to deliver structured, cuisine-aware recipes.
- Architected a dual-mode semantic ingredient substitution engine with an Annoy ANN index and result caching, reducing latency to under **95ms** with high semantic accuracy.
- Designed structured prompt templates for output control and integrated Google NMT via deep-translator to enable one-click recipe translation across languages.

Brain Bio-Signal Classification (EEG) | *PyTorch*, *Transformers*, *Signal Processing* Aug 2025 – Dec 2025

- Developed a hybrid deep learning model (Multi-Scale **CNN** + Gated **Transformer** + **TCN**) for 4-class EEG-based emotion classification on the DEAP dataset for non-verbal patients.
- Applied **median thresholding** via EDA to resolve critical class imbalance, achieving near-uniform distribution across all four emotion classes before training.
- Built an **end-to-end EEG pipeline** (baseline removal, segmentation, normalization) generating 38,400 labelled segments, validated via label correlation and topographic variance analysis.

CreditGuard: Credit Default Risk Prediction | *Risk Prediction*, *SQL*, *Tableau* Aug 2025 – Oct 2025

- Built an ML model predicting credit card default risk for 30,000+ clients by employing **Random Forest** and **XGBoost** models, achieving 81% accuracy and **0.77** ROC-AUC to identify high-risk borrowers.
- Constructed a preprocessing pipeline using log transformation, Standard Scaler, **SMOTE** oversampling, and **Random Under Sampling** to resolve severe class imbalance.
- Designed an **interactive Tableau Risk Monitoring Dashboard** visualizing default probability and model performance across classifiers for risk analysts and stakeholders.

TECHNICAL SKILLS

- **Languages**: Python, SQL, R, C++
- **Generative AI**: RAG, LLM, NLP, Vector Embeddings, Prompt Engineering, Context Engineering
- **Data & Visualization**: Power BI, Tableau, NumPy, Pandas, Matplotlib, Seaborn
- **ML**: Scikit-Learn, Random Forest, XGBoost, CatBoost, Ridge, K-Means, SVR, PCA, SMOTE, PyTorch, CNN, Transformers, Computer Vision, NLP, spaCy
- **Databases**: MySQL, PostgreSQL, SQLAlchemy, AWS DynamoDB, S3, MongoDB, GCP, Azure
- **Tools**: Github, Jira, all-MiniLM-L6-v2, OpenAI Whisper, CUDA, Apache Airflow (DAG design, scheduling, monitoring)